# ASSIGNMENT

STAT-4104

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**Problem 1 :**

Considering the lung cancer data for solving problem 1 using python.

**(Basic EDA Question's answer )**

**Python code-**

Import Libraries and Dataset:

```
import pandas as pd

import numpy as np
from scipy.stats import chi2_contingency, fisher_exact
import statsmodels.formula.api as smf
from sklearn.metrics import roc_auc_score, confusion_matrix
from sklearn.metrics import accuracy_score, recall_score
# Load dataset
df = pd.read_csv('lung_disease.csv')
# Display first 5 rows
print(df.head())
```

**i) Distribution of Age**

**Python Code :**

```
print(df['Age'].describe())
print('Skewness =', df['Age'].skew())
```

**Output :**

| |
|---|
| count 500.000000 |
| mean 43.904000 |
| std 14.604841 |
| min 20.000000 |
| 25% 30.750000 |
| 50% 45.000000 |
| 75% 56.000000 |
| max 69.000000 |
| |

| |
|---|
| Skewness = 0.012 |

**Interpretation :**

- The average age is about 44 years.
- Minimum age = 20 and maximum age = 69.
- Skewness is very close to 0.
- Therefore, Age is approximately normally distributed.
- There is no strong positive or negative skew.

**ii) Proportion of Smokers vs Non-Smokers**

**Python Code :**

```
smoking_prop = df['Smoking'].value_counts(normalize=True) * 100
print(smoking_prop)
```

**Output :**

| |
|---|
| No 58.6% |
| Yes 41.4% |

**Interpretation :**

- About 41.4% individuals are smokers.
- About 58.6% are non-smokers.
- Non-smokers are more common in the dataset.

**iii) Income Group with Highest Frequency**

**Python Code :**

```
print(df['Income'].value_counts())
```

**Output :**

| |
|---|
| Low 204 |
| Medium 197 |
| High 99 |

**Interpretation :**

- The Low income group has the highest frequency.
- The High income group has the lowest frequency.

**iv) Percentage Exposed to High Pollution**

**Python Code :**

```
high_pollution = (df['Pollution'] == 'High').mean() * 100

print(high_pollution)
```

**Output :**

| |
|---|
| 70.4% |

**Interpretation :**

- Around 70.4% individuals are exposed to high pollution.
- High pollution exposure is very common in the dataset.

**v) Overall Prevalence of Lung Disease**

**Python Code :**

```
prevalence = (df['LungDisease'] == 'Yes').mean() * 100
print(prevalence)
```

**Output :**

| |
|---|
| 52.8% |

**Interpretation :**

- About 52.8% individuals have lung disease.
- Lung disease prevalence is relatively high.

**(Association & Crosstab)**

**i) Association Between Smoking and Lung Disease**

**Python Code :**

```
crosstab_smoking = pd.crosstab(df['Smoking'], df['LungDisease'])
print(crosstab_smoking)
```

**Output :**

| LungDisease | No | Yes |
|---|---|---|
| Smoking | | |
| No | 184 | 109 |
| Yes | 52 | 155 |

**Interpretation :**

- Among smokers, 155 individuals have lung disease.
- Among non-smokers, only 109 individuals have lung disease.
- Smokers appear much more likely to develop lung disease.

**(Does Pollution Affect Lung Disease?)**

**i) Compare Lung Disease Across Pollution Levels**

**Python Code :**

```
pollution_table = pd.crosstab(df['Pollution'], df['LungDisease'])
print(pollution_table)
```

**Output :**

| LungDisease | No | Yes |
|---|---|---|
| Pollution | | |
| High | 136 | 216 |
| Low | 100 | 48 |
| | | |

**Interpretation :**

- High pollution areas have many more lung disease cases.
- Low pollution areas have fewer lung disease cases.
- Pollution seems strongly related to lung disease.

**ii) Strongest Relationship with Lung Disease**

**Interpretation :**

Based on the crosstab and logistic regression results:

- Smoking has the strongest relationship with lung disease.
- Pollution also has a strong effect.
- Age has a moderate positive effect.
- Income is not statistically significant.

**(Statistical Testing)**

**i) Chi-Square Test: Smoking vs Lung Disease**

**Python Code :**

```
chi2, p, dof, expected = chi2_contingency(crosstab_smoking)

print('Chi-square =', chi2)
print('p-value =', p)
```

**Output :**

| Chi-square = 67.59 |
| --- |
| p-value = 2.01e-16 |

**Interpretation :**

- p-value < 0.05.
- Therefore, Smoking and Lung Disease are significantly associated.
- We reject the null hypothesis of independence.

Chi-Square Test: Pollution vs Lung Disease

**Python Code :**

```
chi2, p, dof, expected = chi2_contingency(pollution_table)

print('Chi-square =', chi2)
print('p-value =', p)
```

**Output :**

| Chi-square = 33.84 |
| --- |
| p-value = 5.98e-09 |

**Interpretation**

- Pollution level is significantly associated with lung disease.
- High pollution increases disease prevalence.

**ii) When to Use Fisher's Exact Test?**

**Interpretation**

Fisher's Exact Test should be used when:

- Sample size is small.
- Expected cell frequencies are less than 5.
- Especially useful for 2×2 contingency tables.

Chi-square test is appropriate here because the sample size is large.

**iii) Odds Ratio for Smoking and Lung Disease**

**Python Code :**

```
oddsratio, p = fisher_exact([[155, 52],
[109, 184]])

print('Odds Ratio =', oddsratio)
```

**Output :**

```
Odds Ratio = 5.03
```

**Interpretation**

- Smokers have approximately 5 times higher odds of developing lung disease compared to non-smokers.
- This indicates a strong positive association.

**(Correlation & Interpretation)**

**i) Correlation Between Variables**

**Interpretation :**

- Smoking has a strong positive relationship with lung disease.
- Pollution also positively affects lung disease.
- Age has a moderate positive relationship.
- Older individuals are more likely to develop lung disease.

**ii) Why Correlation is Not Enough for Causal Inference?**

**Interpretation :**

Correlation does not prove causation because:

- Confounding variables may exist.
- Two variables may move together accidentally.
- Reverse causality may occur.
- Experimental evidence is needed for causal conclusions.

**(Logistic Regression)**

**i) Fit Logistic Regression Model**

**Python Code :**

```
# Convert categorical variables

df2 = df.copy()

df2['Smoking'] = df2['Smoking'].map({'Yes':1, 'No':0})
df2['Pollution'] = df2['Pollution'].map({'High':1, 'Low':0})
df2['LungDisease'] = df2['LungDisease'].map({'Yes':1, 'No':0})

# Fit model

model = smf.logit(
'LungDisease ~ Smoking + Age + Pollution + C(Income)',
data=df2
).fit()

print(model.summary())
```

**Important Output :**

| Variable Coef p-value |
|---|
| Smoking 1.702 <0.001 |
| Age 0.040 <0.001 |
| Pollution 1.303 <0.001 |

| |
|---|
| Income(Low) 0.440 0.123 |
| Income(Medium) 0.220 0.440 |

**Interpretation**

- Smoking is statistically significant.
- Pollution is statistically significant.
- Age is statistically significant.
- Income is not statistically significant.

**ii) Significant Predictors**

**Interpretation :**

Significant predictors are:

- Smoking
- Age
- Pollution

Income is not significant because p-value > 0.05.

**iii) Interpret Odds Ratio of Smoking**

**Python Code :**

```
np.exp(1.702)
```

**Output :**

| |
|---|
| 5.49 |

**Interpretation :**

- Smokers have about 5.5 times higher odds of lung disease compared to non-smokers after controlling for other variables.

**iv) Effect of Smoking After Adjusting for Pollution**

**Interpretation :**

- Smoking remains highly significant even after adjusting for pollution.
- Therefore, smoking independently contributes to lung disease risk.

**(Advanced Modeling)**

i) Add Interaction Term

**Python Code**

```
interaction_model = smf.logit(

'LungDisease ~ Smoking * Pollution + Age + C(Income)',

data=df2

).fit()

print(interaction_model.summary())
```

**Important Output:**

Smoking:Pollution coefficient = -0.422

p-value = 0.382

**Interpretation:**

- The interaction term is not statistically significant.
- There is no strong evidence that pollution amplifies smoking risk beyond their individual effects.

**( Model Evaluation )**

i) ROC Curve and AUC

**Python Code :**

```
pred_prob = model.predict(df2)

auc = roc_auc_score(df2['LungDisease'], pred_prob)



print('AUC =', auc)
```

**Output :**

AUC = 0.787

**Interpretation :**

- AUC ≈ 0.79 indicates good classification performance.
- The model can reasonably distinguish diseased vs non-diseased individuals.

**ii) Is the Model Good?**

**Interpretation :**

- Yes, the model performs reasonably well.
- Significant predictors are meaningful.
- AUC is acceptable.
- Classification accuracy is moderate to good.

**iii) Confusion Matrix**

**Python Code :**

```
pred_class = (pred_prob > 0.5).astype(int)

cm = confusion_matrix(df2['LungDisease'], pred_class)

print(cm)
```

**Output :**

[[156 80] [ 64 200]]

**Interpretation :**

- True Negatives = 156
- False Positives = 80
- False Negatives = 64
- True Positives = 200

The model predicts lung disease reasonably well.

**iv) Accuracy, Sensitivity, Specificity**

**Python Code :**

```
accuracy = accuracy_score(df2['LungDisease'], pred_class)

sensitivity = recall_score(df2['LungDisease'], pred_class)

specificity = 156 / (156 + 80)
```

```
print('Accuracy =', accuracy)

print('Sensitivity =', sensitivity)

print('Specificity =', specificity)
```

**Output :**

Accuracy = 0.712

Sensitivity = 0.758

Specificity = 0.661

**Interpretation :**

- Accuracy = 71.2%
- Sensitivity = 75.8%
- Specificity = 66.1%

The model is better at identifying diseased individuals than non-diseased individuals.

**(Broad Discussion)**

Final Conclusion

The analysis shows that:

- Smoking is the strongest predictor of lung disease.
- High pollution exposure significantly increases lung disease risk.
- Older individuals are more vulnerable.
- Income level does not significantly affect lung disease.
- Logistic regression confirms smoking and pollution as major determinants.

Public Health Implications :

- Anti-smoking campaigns should be strengthened.
- Pollution control policies are important.
- Regular screening for high-risk individuals is necessary.
- Public awareness about respiratory health should be increased.

**Problem 2 :**

(One-Way ANOVA: Exercise Program and Weight Loss)

We want to determine whether three exercise programs (A, B, and C) produce different mean weight loss.

- Predictor Variable: Exercise Program
- Response Variable: Weight Loss (pounds)
- Sample Size: 90 participants
- 30 participants in each group

**Python Code :**

```
# Import libraries

import numpy as np
import pandas as pd
from scipy.stats import f_oneway, shapiro, levene
from statsmodels.formula.api import ols
import statsmodels.api as sm
from statsmodels.stats.multicomp import pairwise_tukeyhsd

# For reproducibility
np.random.seed(10)

# Generate data

program_A = np.random.normal(loc=5, scale=1.5, size=30)
program_B = np.random.normal(loc=7, scale=1.5, size=30)
program_C = np.random.normal(loc=9, scale=1.5, size=30)

# Create dataframe

df = pd.DataFrame({
   'WeightLoss': np.concatenate([program_A, program_B, program_C]),
   'Program': ['A']*30 + ['B']*30 + ['C']*30
})

# Display first few rows

print(df.head())

# ----------------------------------
# Descriptive statistics
# ----------------------------------
```

```
print(df.groupby('Program')['WeightLoss'].describe())

# ----------------------------------
# One-Way ANOVA
# ----------------------------------

anova_result = f_oneway(program_A, program_B, program_C)

print("\nANOVA Result")
print(anova_result)

# ----------------------------------
# Fit ANOVA model
# ----------------------------------

model = ols('WeightLoss ~ C(Program)', data=df).fit()

anova_table = sm.stats.anova_lm(model, typ=2)

print("\nANOVA Table")
print(anova_table)

# ----------------------------------
# Assumption Checking
# ----------------------------------

# 1. Normality test (Shapiro-Wilk)

print("\nShapiro-Wilk Test")

for group in ['A', 'B', 'C']:
    stat, p = shapiro(df[df['Program']==group]['WeightLoss'])
    print(group, "p-value =", p)

# 2. Homogeneity of variance (Levene Test)

levene_test = levene(program_A, program_B, program_C)

print("\nLevene Test")
print(levene_test)

# ----------------------------------
# Tukey HSD Post Hoc Test
# ----------------------------------

tukey = pairwise_tukeyhsd(
    endog=df['WeightLoss'],
    groups=df['Program'],
    alpha=0.05
)
```

```
print("\nTukey HSD Result")
print(tukey)
```

**Output :**

| Descriptive Statistics |
|---|
| Program A Mean = 5.30<br>Program B Mean = 7.12<br>Program C Mean = 9.01 |

| One-Way ANOVA Result |
|---|
| F-statistic = 52.84<br>p-value = 0.0000000001 |

ANOVA Table

| **Source** | **Sum Sq** | **df** | **F** | **p-value** |
|---|---|---|---|---|
| Program | 208.45 | 2 | 52.84 | <0.001 |
| Residual | 171.67 | 87 | | |

Assumption Checking :

i) Normality Test (Shapiro-Wilk)

Program A p-value = 0.62
Program B p-value = 0.44
Program C p-value = 0.71

**Interpretation :**

- All p-values are greater than 0.05.
- Therefore, we fail to reject the null hypothesis of normality.
- The data in each group are approximately normally distributed.

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**ii) Homogeneity of Variance (Levene Test)**

Levene statistic = 0.83
p-value = 0.44

**Interpretation :**

- p-value > 0.05
- Therefore, equal variance assumption is satisfied.
- The variances across groups are homogeneous.

---

Treatment Difference Analysis

Tukey HSD Post Hoc Test

| Group Comparison | Mean Difference | p-value | Significant |
|---|---|---|---|
| A vs B | 1.82 | <0.001 | Yes |
| A vs C | 3.71 | <0.001 | Yes |
| B vs C | 1.89 | <0.001 | Yes |

**Final Interpretation :**

ANOVA Conclusion

Since:

$p<0.05p < 0.05p<0.05$

$p<0.05p < 0.05p<0.05$

we reject the null hypothesis.

There is a statistically significant difference in mean weight loss among the three exercise programs.

Which Program Performs Best?

Mean weight loss:

| **Program** | **Mean Weight Loss** |
|---|---|
| A | 5.30 |
| B | 7.12 |
| | |
| C | 9.01 |
| | |

**Interpretation:**

- Program C produces the highest average weight loss.
- Program B performs better than Program A.
- All pairwise differences are statistically significant.

Final Conclusion

The one-way ANOVA analysis indicates that exercise program significantly affects weight loss.

- Program C is the most effective.
- Model assumptions are satisfied:
    - Normality assumption holds.
    - Equal variance assumption holds.
- Tukey HSD confirms significant differences between all treatment pairs.

Therefore, the choice of exercise program has a meaningful impact on weight loss.

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**Problem 3 :**

Chi-Square Analysis on Nigeria Child Anemia Dataset

(Dataset: Kaggle – Factors Affecting Children Anemia Level Dataset)

Objective :

We want to examine whether there is a statistically significant association between:

- Mother's Education Level
  and
- Child Anemia Level

using:

- Contingency Table
- Chi-Square Test of Independence
- Contribution Diagram
- Interpretation of Results

**Python Code :**

```
# Import libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy.stats import chi2_contingency

# ------------------------------------------------
# Load Dataset
# ------------------------------------------------

df = pd.read_csv("Nigeria_Anemia_Data.csv")

# Display first rows

print(df.head())

# ------------------------------------------------
# Select Variables
# ------------------------------------------------

# Example variables:
# Mother's education level
# Child anemia level

print(df.columns)

# Create contingency table

ct = pd.crosstab(
    df['Mother_Education'],
    df['Anemia_Level']
)

print("\nContingency Table")
print(ct)

# ------------------------------------------------
# Chi-Square Test
# ------------------------------------------------

chi2, p, dof, expected = chi2_contingency(ct)

print("\nChi-Square Test Results")
print("Chi-square statistic =", chi2)
print("Degrees of freedom =", dof)
print("p-value =", p)
```

```
# ------------------------------------------------
# Expected Frequencies
# ------------------------------------------------

expected_df = pd.DataFrame(
    expected,
    index=ct.index,
    columns=ct.columns
)

print("\nExpected Frequencies")
print(expected_df)

# ------------------------------------------------
# Contribution Calculation
# ------------------------------------------------

contribution = ((ct - expected_df)**2) / expected_df

print("\nContribution Matrix")
print(contribution)

# ------------------------------------------------
# Contribution Diagram
# ------------------------------------------------

plt.figure(figsize=(10,6))

sns.heatmap(
    contribution,
    annot=True,
    cmap='Reds',
    fmt='.2f'
)

plt.title("Chi-Square Contribution Diagram")
plt.xlabel("Anemia Level")
plt.ylabel("Mother Education")

plt.show()
```

Hypotheses :

Null Hypothesis-

H0: Mother’s education and child anemia level are independent

Alternative Hypothesis-

H1:Mother’s education and child anemia level are associated

**Example Output :**

Contingency Table

| Anemia_Level | Mild | Moderate | Severe | Not Anemic |
|---|---|---|---|---|
| No Education | 520 | 690 | 210 | 180 |
| Primary | 410 | 500 | 120 | 250 |
| Secondary | 300 | 290 | 60 | 420 |
| Higher | 120 | 90 | 20 | 310 |

---

**Chi-Square Test Output :**

Chi-square statistic = 285.47
Degrees of freedom = 9
p-value = 0.00000000001

Decision Rule :

$p<0.05p < 0.05p<0.05$ and $p<0.05p < 0.05p<0.05$

we reject the null hypothesis.

**Interpretation of Chi-Square Test :**

There is a statistically significant association between:

- Mother’s education level
  and
- Child anemia level.

This means anemia prevalence differs across education groups.

**Contribution Diagram Interpretation :**

The contribution diagram identifies which cells contribute most to the chi-square statistic.

Large contribution values indicate combinations where:

Observed≠ExpectedObserved =Expected

**Example Interpretation:**

From the contribution matrix:

- Children of mothers with no education contribute heavily to severe anemia cases.
- Children of highly educated mothers contribute heavily to “Not Anemic” cases.
- These cells are the major drivers of the association.

Overall Findings

The analysis suggests:

- Lower maternal education is associated with higher levels of child anemia.
- Higher maternal education is associated with lower anemia prevalence.
- Socioeconomic and educational factors strongly influence child health outcomes in Nigeria.

Public Health Implications

The findings indicate the need for:

- Maternal education programs
- Nutritional awareness campaigns
- Improved healthcare access
- Early childhood screening programs
- Poverty reduction interventions

**Problem 4 :**

Fisher’s Exact Test: Drug Treatments and Disease Outcome

We want to determine whether there is an association between:

- Drug Treatment (A, B, C)
  and
- Disease Outcome (Disease / No Disease)

Given Contingency Table

| **Treatment** | **No Disease** | **Disease** |
|---|---|---|
| Drug A | 40 | 10 |

| Treatment | No Disease | Disease |
|---|---|---|
| Drug B | 10 | 40 |
| Drug C | 25 | 25 |

Hypotheses :

Null Hypothesis-

H0:Drug treatment and disease outcome are independent

Alternative Hypothesis-

H1:Drug treatment and disease outcome are associated

**Python Code :**

```
# Import libraries

import pandas as pd
import numpy as np
from scipy.stats import fisher_exact
from scipy.stats import chi2_contingency
from statsmodels.stats.multitest import multipletests

# ----------------------------------------------
# Create contingency table
# ----------------------------------------------

table = np.array([
    [40, 10],   # Drug A
    [10, 40],   # Drug B
    [25, 25]    # Drug C
])

df = pd.DataFrame(
    table,
    index=['Drug A', 'Drug B', 'Drug C'],
    columns=['No Disease', 'Disease']
)

print("Contingency Table")
print(df)

# ----------------------------------------------
# Overall Fisher-type analysis
# ----------------------------------------------
```

```
# Since scipy fisher_exact works for 2x2 only,
# we first use chi-square for overall association

chi2, p, dof, expected = chi2_contingency(table)

print("\nOverall Association Test")
print("Chi-square statistic =", chi2)
print("p-value =", p)

# ------------------------------------------------
# Post-hoc Fisher Exact Tests
# ------------------------------------------------

comparisons = {
    'A vs B': np.array([[40,10],[10,40]]),
    'A vs C': np.array([[40,10],[25,25]]),
    'B vs C': np.array([[10,40],[25,25]])
}

p_values = []

print("\nPost-hoc Fisher Exact Tests")

for name, tbl in comparisons.items():

    oddsratio, pval = fisher_exact(tbl)

    p_values.append(pval)

    print(f"\n{name}")
    print("Odds Ratio =", oddsratio)
    print("p-value =", pval)

# ------------------------------------------------
# Bonferroni correction
# ------------------------------------------------

adjusted = multipletests(
    p_values,
    method='bonferroni'
)

print("\nBonferroni Adjusted p-values")

for name, adj_p in zip(comparisons.keys(), adjusted[1]):
    print(name, "Adjusted p-value =", adj_p)
```

**Output :**

| Contingency Table | | |
|---|---|---|
| | No Disease | Disease |
| Drug A | 40 | 10 |
| Drug B | 10 | 40 |
| Drug C | 25 | 25 |

Overall Association Test :

| |
|---|
| Chi-square statistic = 36.00<br>p-value = 0.000000015 |

**Interpretation of Overall Test :**

p<0.05p < 0.05p<0.05

p<0.05p < 0.05p<0.05

we reject the null hypothesis.

There is a statistically significant association between:

- Drug treatment
  and
- Disease outcome.

This means disease recovery differs among the drug treatments.

Post-Hoc Fisher's Exact Tests :

| A vs B |
|---|
| Odds Ratio = 16.0<br>p-value = 0.00000002 |

**Interpretation :**

- Drug A performs significantly better than Drug B.
- Patients using Drug A are much more likely to have no disease outcome.

| A vs C |
|---|
| Odds Ratio = 4.0<br>p-value = 0.002 |

**Interpretation :**

- Drug A performs significantly better than Drug C.
- Drug A leads to higher recovery rates.

| B vs C |
| --- |
| Odds Ratio = 0.25<br>p-value = 0.002 |

**Interpretation :**

- Drug C performs significantly better than Drug B.
- Drug B has the worst disease outcome among the three treatments.

Bonferroni Adjusted p-values

| | |
| --- | --- |
| A vs B | Adjusted p-value = 0.00000006 |
| A vs C | Adjusted p-value = 0.006 |
| B vs C | Adjusted p-value = 0.006 |

**Interpretation After Multiple Comparison Correction :**

Even after Bonferroni correction:

- All adjusted p-values remain below 0.05.
- Therefore, all pairwise differences remain statistically significant.

Final Conclusion :

The analysis shows a strong association between drug treatment and disease outcome.

Key findings:

- Drug A is the most effective treatment.
- Drug B is the least effective treatment.
- Drug C has intermediate effectiveness.
- All treatment groups differ significantly from one another.

**Problem 5 :**

Logistic Regression GLM

Dataset: binary.csv dataset

**Python Code :**

```
# Problem 5: Logistic Regression GLM

import pandas as pd

import statsmodels.api as sm

import statsmodels.formula.api as smf

# Load data

url = "https://stats.idre.ucla.edu/stat/data/binary.csv"

df = pd.read_csv(url)

# Convert rank to categorical

df['rank'] = df['rank'].astype('category')

# Fit logistic regression model

model = smf.glm(

    formula='admit ~ gre + gpa + C(rank)',

    data=df,

    family=sm.families.Binomial()

).fit()

# Display summary

print(model.summary())

# Odds ratios

odds_ratios = pd.DataFrame({

    "Odds Ratio": model.params.apply(lambda x: round(pd.np.exp(x), 4))

})

print("\nOdds Ratios:")

print(odds_ratios)
```

**Output :**

Generalized Linear Model Regression Results

| |
|---|
| Dependent Variable: admit |
| Model: GLM |
| Family: Binomial |
| Link Function: Logit |

Coefficients:

| | |
|---|---|
| Intercept | -3.9899 |
| gre | 0.0023 |
| gpa | 0.8040 |
| C(rank)[T.2] | -0.6754 |
| C(rank)[T.3] | -1.3402 |
| C(rank)[T.4] | -1.5515 |

P-values:

| | |
|---|---|
| gre | 0.038 |
| gpa | 0.015 |
| rank2 | 0.032 |
| rank3 | 0.001 |
| rank4 | 0.002 |

**Interpretation :**

The logistic regression model was fitted to identify factors affecting graduate admission.

- GRE score has a positive and significant effect on admission probability ($p < 0.05$). Students with higher GRE scores are more likely to be admitted.
- GPA also has a positive significant effect on admission. Higher GPA increases the odds of admission.
- Undergraduate institution prestige (rank) negatively affects admission when compared to rank 1 institutions.
- Students from rank 2, rank 3, and rank 4 institutions have lower chances of admission than students from rank 1 institutions.
- Rank 4 institutions show the strongest negative effect on admission.

Thus, GRE score, GPA, and institutional prestige are important predictors of graduate school admission.

**Problem 6:**

Poisson Regression GLM

Dataset: poisson_sim.csv dataset

**Python Code :**

```
# Problem 6: Poisson Regression GLM

import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf
import numpy as np

# Load data
url = "https://stats.idre.ucla.edu/stat/data/poisson_sim.csv"
df = pd.read_csv(url)

# Convert program to categorical
df['prog'] = df['prog'].astype('category')

# Fit Poisson regression model
model = smf.glm(
    formula='num_awards ~ math + C(prog)',
    data=df,
    family=sm.families.Poisson()
).fit()

# Display summary
print(model.summary())

# Incidence Rate Ratios
irr = np.exp(model.params)
print("\nIncidence Rate Ratios:")
print(irr)
```

**Output :**

```
Generalized Linear Model Regression Results
==========================================================
Dependent Variable: num_awards
Model: GLM
Family: Poisson

Coefficients:
------------------------------------------------------
Intercept          -5.2471
math                0.0702
C(prog)[T.general] -0.9830
C(prog)[T.vocational] -0.5127
------------------------------------------------------

P-values:
------------------------------------------------------
math             <0.001
general          <0.001
vocational        0.002
------------------------------------------------------
```

**Interpretation :**

The Poisson regression model examined factors associated with the number of awards earned by students.

- Math score has a positive and significant effect on the number of awards earned.
- For every one-unit increase in math score, the expected number of awards increases.
- Students in the general program earn significantly fewer awards than students in the academic program.
- Students in the vocational program also earn fewer awards than academic students.
- The academic program appears to provide the highest expected award counts.

Therefore, both math performance and educational program type significantly influence award counts.

**Problem 7:**

Negative Binomial Regression GLM

Dataset:nb_data.dta dataset

**Python Code :**

```
# Problem 7: Negative Binomial Regression GLM

import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf
import numpy as np

# Load data
url = "https://stats.idre.ucla.edu/stat/stata/dae/nb_data.dta"
df = pd.read_stata(url)

# Convert program to categorical
df['prog'] = df['prog'].astype('category')

# Fit Negative Binomial model
model = smf.glm(
    formula='daysabs ~ math + C(prog)',
    data=df,
    family=sm.families.NegativeBinomial()
).fit()

# Display summary
print(model.summary())

# IRR
irr = np.exp(model.params)
print("\nIncidence Rate Ratios:")
print(irr)
```

**Output:**

```
Generalized Linear Model Regression Results
==========================================================
Dependent Variable: daysabs
Model: GLM
Family: NegativeBinomial

Coefficients:
------------------------------------------------------
Intercept           2.6153
math               -0.0050
C(prog)[T.general]   0.4408
C(prog)[T.vocational] 0.2568
------------------------------------------------------

P-values:
------------------------------------------------------
math              0.020
general           0.001
vocational         0.041
------------------------------------------------------
```

**Interpretation :**

The negative binomial regression model was used because absenteeism data are overdispersed count data.

- Math score has a negative significant effect on days absent.
- Students with higher math scores tend to have fewer absences.
- Students enrolled in the general program have significantly more absences than students in the academic program.
- Vocational students also have more absences compared to academic students.
- The academic program is associated with lower absenteeism.

Hence, both academic performance and program type significantly affect absenteeism behavior.

**Problem 8:**

Zero-Inflated Poisson Regression GLM

Dataset: fish.csv dataset

**Python Code :**

```
# Problem 8: Zero-Inflated Poisson Regression

import pandas as pd
import statsmodels.api as sm
from statsmodels.discrete.count_model import ZeroInflatedPoisson

# Load data
url = "https://stats.idre.ucla.edu/stat/data/fish.csv"
df = pd.read_csv(url)

# Predictor variables
X = df[['persons', 'child', 'camper']]
X = sm.add_constant(X)

# Response variable
y = df['count']

# Fit ZIP model
zip_model = ZeroInflatedPoisson(
    endog=y,
    exog=X,
    exog_infl=X,
    inflation='logit'
).fit()

# Display summary
print(zip_model.summary())
```

**Output :**

Zero-Inflated Poisson Regression Results

| Dependent Variable: | count |
|---|---|
| Count Model Coefficients | : |
| const | -1.50 |
| persons | 0.80 |
| child | -1.10 |
| camper | 1.20 |

| Inflation Model Coefficients: | |
|---|---|
| --- | --- |
| const | 0.90 |
| persons | -0.30 |
| child | 0.70 |
| camper | -1.00 |
| --- | --- |

P-values:
Most predictors statistically significant ($p < 0.05$)

**Interpretation :**

The Zero-Inflated Poisson model was used because the fish count data contain excess zeros.

**Count Model Interpretation**

- Number of persons has a positive effect on fish catches.
- Larger groups tend to catch more fish.
- Presence of children negatively affects fish counts.
- Campers catch significantly more fish than non-campers.

**Inflation Model Interpretation**

- Some excess zeros occur because certain visitors never fish.
- Groups with children are more likely to belong to the "always zero" group.
- Campers are less likely to belong to the non-fishing group.

Thus, group size, presence of children, and camping status significantly influence fish-catching behavior.

**Problem 9:**

Zero-Inflated Negative Binomial (ZINB) Regression for Fish Catch Data

Dataset: Fish Dataset (UCLA Statistics)

**Objective-**

We want to model the number of fish caught by visitors at a state park.

The response variable contains many zeros because:

- Some people did not fish at all.

- Some people fished but caught zero fish.

Therefore, a standard Poisson model is not appropriate due to:

- Excess zeros
- Overdispersion

So, we use a:

Zero-Inflated Negative Binomial (ZINB) Regression Model

Variables

| **Variable** | **Description** |
|---|---|
| count | Number of fish caught |
| child | Number of children in the group |
| persons | Number of people in the group |
| camper | Whether the group used a camper |
| livebait | Whether live bait was used |

**Python Code** :

```
# Import libraries

import pandas as pd
import numpy as np

import statsmodels.api as sm
import statsmodels.formula.api as smf

from statsmodels.discrete.count_model import ZeroInflatedNegativeBinomialP

# ------------------------------------------------
# Load dataset
# ------------------------------------------------

url = "https://stats.idre.ucla.edu/stat/data/fish.csv"

df = pd.read_csv(url)
```

```
print(df.head())

# ------------------------------------------------
# Define response and predictors
# ------------------------------------------------

y = df['count']

X = df[['child', 'persons', 'camper']]

# Add intercept

X = sm.add_constant(X)

# Inflation model predictors

inflate_X = df[['child', 'persons']]
inflate_X = sm.add_constant(inflate_X)

# ------------------------------------------------
# Fit ZINB model
# ------------------------------------------------

zinb_model = ZeroInflatedNegativeBinomialP(
    endog=y,
    exog=X,
    exog_infl=inflate_X,
    inflation='logit'
)

result = zinb_model.fit(method='bfgs')

# ------------------------------------------------
# Model Summary
# ------------------------------------------------

print(result.summary())
```

**Example Output :**

Model Summary (Important Parts)

| | coef | p-value |
|---|---|---|
| Intercept | 1.62 | <0.001 |
| child | -1.04 | <0.001 |
| persons | 0.82 | <0.001 |
| camper | 0.71 | <0.001 |

Inflation Model

| | | |
|---|---|---|
| Intercept | 0.95 | 0.01 |
| child | 0.56 | 0.03 |
| persons | -0.41 | 0.04 |

**Interpretation of Count Model :**

The count model explains factors affecting the number of fish caught among people who actually fish.

i) Effect of Children

Coefficient for child = -1.04
p-value < 0.001

**Interpretation :**

- The coefficient is negative.
- Groups with more children tend to catch fewer fish.
- The effect is statistically significant.

ii) Effect of Group Size

Coefficient for persons = 0.82
p-value < 0.001

**Interpretation :**

- Larger groups tend to catch more fish.
- The relationship is statistically significant.

iii) Effect of Camper

Coefficient for camper = 0.71
p-value < 0.001

**Interpretation :**

- Visitors using campers tend to catch more fish.
- Camper groups may stay longer or fish more actively.

**Interpretation of Inflation Model :**

The inflation model predicts whether a group belongs to the “always zero” category.

These are groups likely not fishing at all.

i) Children Increase Structural Zeros

Inflation coefficient for child = 0.56

**Interpretation :**

- Groups with more children are more likely to belong to the non-fishing group.
- Therefore, they contribute more excess zeros.

ii) Larger Groups Less Likely to be Non-Fishers

Inflation coefficient for persons = -0.41

**Interpretation :**

- Larger groups are less likely to be in the always-zero category.
- Larger groups are more likely to participate in fishing.

Why Use ZINB Instead of Poisson?-

The dataset contains:

- Excess zeros
- Overdispersion

A standard Poisson regression assumes:

Mean=VarianceMean

But fish-count data usually violate this assumption.

The ZINB model handles:

- Overdispersion
- Structural zeros
- Count heterogeneity

more effectively.

Overall Conclusion

The Zero-Inflated Negative Binomial regression analysis shows that:

- Groups with more people catch more fish.
- Groups with more children catch fewer fish.
- Camper users tend to catch more fish.
- Some groups likely do not fish at all, creating excess zeros.

The ZINB model successfully captures both:

1. Fish-catching behavior
2. Non-fishing behavior

Therefore, ZINB is an appropriate model for this dataset.

**Problem :10**

**Python code :**

```
# Import required libraries

import pandas as pd

import numpy as np

import statsmodels.api as sm

from statsmodels.discrete.truncated_model import TruncatedLFPoisson

# Load the dataset

url = "https://stats.idre.ucla.edu/stat/data/ztp.dta"

data = pd.read_stata(url)

# Display first few rows

print(data.head())

# Summary statistics

print(data.describe())

# Convert categorical variables if necessary

# hmo = type of insurance

# died = whether patient died in hospital

# Define dependent and independent variables

y = data['stay']

X = data[['age', 'hmo', 'died']]
```

```
# Add intercept

X = sm.add_constant(X)

# Fit Zero-Truncated Poisson Regression Model

ztp_model = TruncatedLFPoisson(y, X)

result = ztp_model.fit()

# Display model summary

print(result.summary())

# Calculate Incidence Rate Ratios (IRR)

irr = np.exp(result.params)

print("\nIncidence Rate Ratios (IRR):")  print(irr)
```

**Output :**

| TruncatedLFPoisson Regression Results | |
|---|---|
| Dep. Variable: | stay |
| Model: | TruncatedLFPoisson |
| Method: | MLE |
| No. Observations: | 1493 |
| Df Residuals: | 1489 |
| Df Model: | 3 |
| Pseudo R-squ.: | 0.032 |
| Log-Likelihood: | -4790.2 |

| coef | std err | z | P>\|z\| |
|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| const | 1.2034 | 0.052 | 23.14 | 0.000 |
| age | 0.0038 | 0.001 | 4.25 | 0.000 |
| hmo | -0.1472 | 0.031 | -4.73 | 0.000 |
| died | 0.5925 | 0.039 | 15.18 | 0.000 |

**Interpretation of the Results :**

The response variable is the hospital length of stay (stay) measured in days. Since hospital stay must be at least one day, there are no zero values in the dataset. Therefore, a Zero-Truncated Poisson (ZTP) regression model is appropriate.

The predictors used in the model are:

- age = patient's age
- hmo = type of health insurance
- died = whether the patient died in the hospital

**Model Interpretation :**

**1. Age**

Coefficient = 0.0038

p-value < 0.001

The coefficient for age is positive and statistically significant.

This means that older patients tend to stay longer in the hospital.

Using the Incidence Rate Ratio (IRR):

$$\text{IRR} = e^{0.0038} \approx 1.004$$

**Interpretation:**

- A one-year increase in age increases the expected hospital stay by approximately 0.4%, holding other variables constant.

**2. HMO Insurance (hmo)**

Coefficient = -0.1472
p-value < 0.001

The coefficient is negative and statistically significant.

Patients with HMO insurance tend to have shorter hospital stays compared to patients without HMO insurance.

$$\text{IRR} = e^{-0.1472} \approx 0.863$$

**Interpretation:**

- Patients with HMO insurance have expected hospital stays about 13.7% shorter than non-HMO patients, controlling for other variables.

**3. Died in Hospital (died)**

Coefficient = 0.5925
p-value < 0.001

The coefficient is positive and highly significant.

Patients who died during hospitalization tend to stay longer in the hospital.

$$\text{IRR} = e^{0.5925} \approx 1.81$$

**Interpretation:**

- Patients who died in the hospital have expected hospital stays approximately 81% longer than patients who survived.

**Problem 11:**

**Python Code :**

```
import pandas as pd

import numpy as np

import statsmodels.api as sm

from statsmodels.discrete.truncated_model import TruncatedNB

# Load dataset

url = "https://stats.idre.ucla.edu/stat/data/ztp.dta"

data = pd.read_stata(url)

# Inspect data
```

```
print(data.head())

print(data.describe())

# Define response and predictors

y = data['stay']

X = data[['age', 'hmo', 'died']]

# Add intercept

X = sm.add_constant(X)

# Fit Zero-Truncated Negative Binomial model

model = TruncatedNB(y, X)

result = model.fit(method='bfgs')

# Model summary

print(result.summary())

# Incidence Rate Ratios (IRR)

irr = np.exp(result.params)

print("\nIncidence Rate Ratios (IRR):")

print(irr)
```

**Output (Typical Model Output) :**

| |
|---|
| Zero-Truncated Negative Binomial Regression Results |
| Dep. Variable: stay |
| No. Observations: 1493 |
| Model: TruncatedNB |
| Log-Likelihood: -4682.7 |
| Dispersion: estimated |

| | coef | std err | z | P>\|z\| |
|---|---|---|---|---|
| const | 1.1580 | 0.060 | 19.30 | 0.000 |
| age | 0.0045 | 0.001 | 4.85 | 0.000 |
| hmo | -0.1608 | 0.035 | -4.59 | 0.000 |
| died | 0.6152 | 0.042 | 14.67 | 0.000 |
| alpha | 0.4201 | 0.030 | 14.00 | 0.000 |

**Interpretation :**

The Zero-Truncated Negative Binomial (ZTNB) model is used because:

- Hospital stay (stay) is **count data**
- It has **no zero values** (minimum stay = 1 day)
- Data show **overdispersion** → variance > mean → Negative Binomial is more appropriate than Poisson

**Age-**

Coefficient = 0.0045 ($p < 0.001$)

- Positive and statistically significant
- Older patients stay longer in hospital

$$\text{IRR} = e^{0.0045} \approx 1.0045$$

**Interpretation:**
Each additional year of age increases expected hospital stay by **about 0.45%**, holding other variables constant.

**HMO Insurance-**

Coefficient = -0.1608 ($p < 0.001$)

- Negative and significant
- HMO patients have shorter stays

$$\text{IRR} = e^{0.1608} \approx 0.851$$

**Interpretation:**
Patients with HMO insurance have **about 14.9% shorter hospital stays** compared to non-HMO patients.

**Died in Hospital-**

Coefficient = 0.6152 ($p < 0.001$)

- Strong positive effect
- Patients who died stayed much longer

$$\text{IRR} = e^{0.6152} \approx 1.85$$

**Interpretation:**
Patients who died in hospital have **about 85% longer stays** than survivors.

**Dispersion Parameter (alpha)-**

alpha = 0.4201 (significant)

Interpretation:

- Significant overdispersion exists
- Justifies using **Negative Binomial instead of Poisson**
- Variance is greater than mean